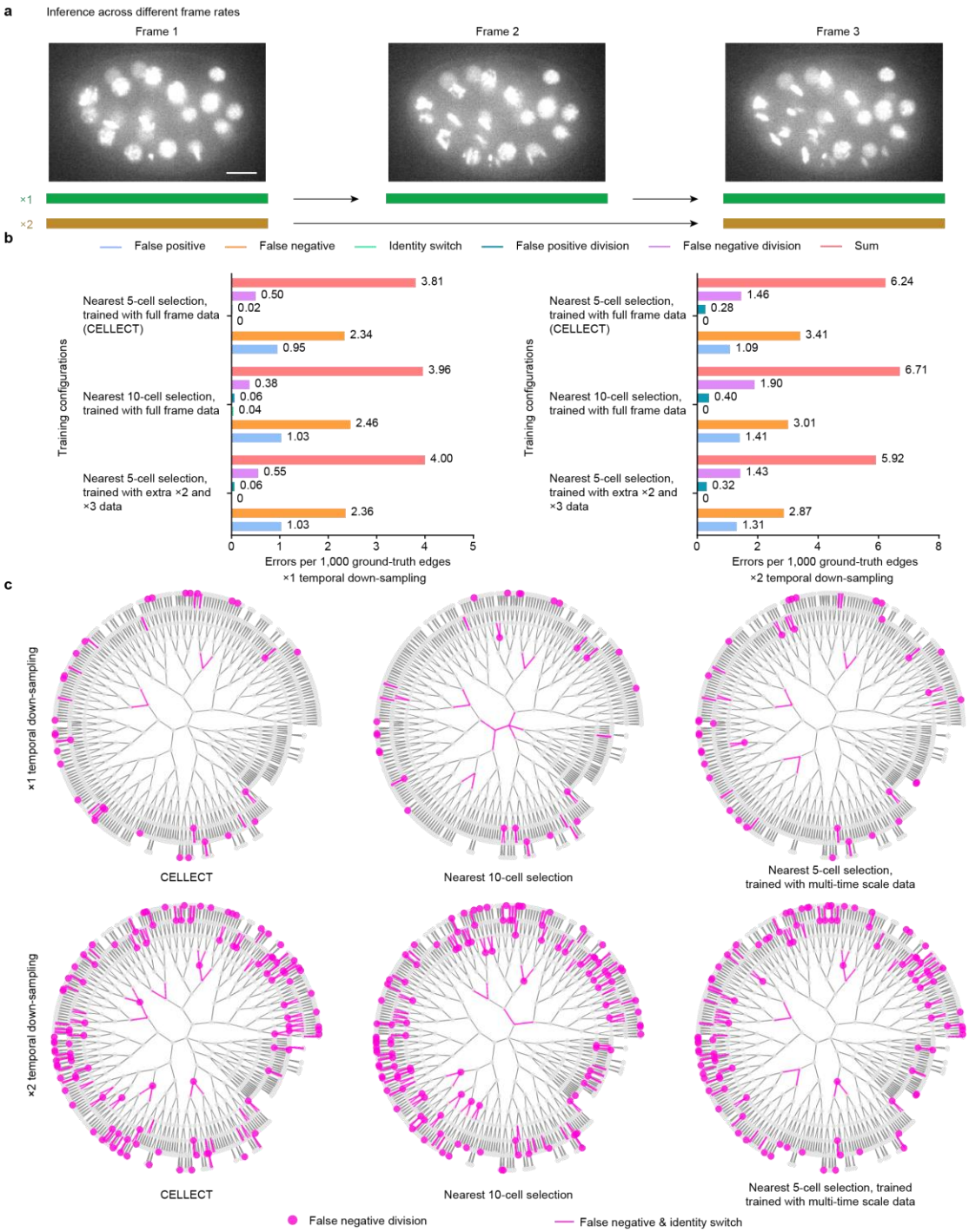


Supplementary Figure 12 | Influence of imaging frame rate on tracking performance under different training dataset and candidate numbers used. a, Illustration of the 2 temporal resolutions used in the inference process: ×1 (75 s) and ×2 (150 s) down-sampling. **b**, Tracking performance of 3 different configurations for different temporal down-sampling in inference: the default CELLECT configuration, model using 10 candidate cells, and model using 5 candidate cells but trained with different temporal resolutions (randomly incorporating dataset with ×1, ×2 and ×3 temporal down-sampling). All configurations show comparable performance under ×1 temporal down-sampling. However, under ×2 temporal down-sampling, models trained with an expanded candidate pool or mixed-temporal-resolution data exhibit reduced false negative (FN) rates, indicating improved robustness to lower temporal resolution. **c**, Circular lineage diagrams